Exertional Heatstroke

Section 1: Case Summary

Scenario Title:	Exertional Heatstroke
Keywords:	Heat related illness, exertional heat stroke, hyperthermia
Brief Description of Case:	In this case, a 58 y/o male experiences exertional heat stroke on a hot summer day during a heat wave. The team must actively cool the patient while working the patient up for other causes. Ultimately, the patient will require admission to hospital for ongoing treatment.

Goals and Objectives	
Educational Goal:	Diagnose and treat exertional heat stroke
Objectives: (Medical and CRM)	1. Maintain a broad differential diagnosis for the hyperthermic and agitated patient 2. Recognize and treat exertional heat stroke 3. Maintain situational awareness, closed loop communication, and role delegation
EPAs Assessed:	

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners	☒ Senior Learners	☒ Staff	
	☒ Physicians	☒ Nurses	☒ RTs	☒ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☒ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors:			
	Sim Actors:			
	Sim Techs:			

Scenario Development	
Date of Development:	June 13, 2023
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Chet			Age: 58	Gender: M	Weight: 85kg
Presenting complaint: Altered LoC					
Temp: 41.7 C	HR: 135	BP: 90/58	RR: 30	O_2Sat: 96%	FiO_2: Room air
Cap glucose: 5.3 mmol/L			GCS: 13 (E 4 V 4 M 5)		
Triage note: EMS has brought in a 58 y/o male who was found "acting strangely". With EMS, the patient has been confused and slurring his speech. He was out running on a hot day. His initial vital signs are: HR 135, RR 30, T 41.7C (can omit and only give if asked), BP 90/58, O2 sat 96% on room air.					
Allergies: NKDA					
Past Medical History: HTN			Current Medications: Ramipril		

Section 2B: Extra Patient Information

A. Further History
The patient's wife was contacted based on ID found in the patient's wallet. The patient had consumed a couple of beers this while gardening all afternoon. An hour or two later told his wife he was heading out for a jog. Of note, it is a particularly hot day with temperatures approximately 38 C during a heat wave.

B. Physical Exam	
List any pertinent positive and negative findings	
Cardio: Normal heart sounds, bounding pulses	Neuro: confused, combative, no focal deficits
Resp: GAEB	Head & Neck: No signs of trauma
Abdo: soft, non tender	MSK/skin: Dry mucous membranes, hot to touch
Other:	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin *(specify type and whether infant/child/adult)*
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Cold water immersion equipment IV fluids Crash cart
C. Required Medications
D. Moulage
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 145 BP: 90/58 RR: 30 O_2SAT: 96 % T: 41.7 ^{o}C GCS: 13	Confused, slurred speech, uncooperative	Expected Learner Actions ☐ IV access ☐ Ensure monitors applied ☐ Ask for core temperature ☐ Cap glucose ☐ Physical exam + history ☐ Remove clothing ☐ Begin cooled fluids IV (4C) ☐ labs, ECG, +/- CT head	Modifiers - Can opt to not tell learners temperature, and only give when asked Triggers *For progression to next state* - Once work up ordered → 2. Initiate treatment	
2. Initiate Treatment	Same as 1st state	Expected Learner Actions ☐ Consider empiric abx for infection post blood cultures ☐ Give fluids bolus for hypotension ☐ Call for cold water immersion ☐ Fan and mist cold water ☐ Ice packs to all skin surfaces ☐ Reduce room temperature ☐ Benzos or dexmedetomidine for agitation/shivering ☐ Insert Foley for urine output and bladder temperature probe ☐ Cooling blanket	Modifiers - antipyretics are not indicated Triggers - Once treatment initiated → 3. Investigations return	
3. Investigations return T 40.1 HR 110 BP 104/72		Expected Learner Actions ☐ Review labs/ECG ☐ Continue fluid resuscitation ☐ Vocalize plan for cooling targets (rapidly to <40, and	Modifiers Triggers	

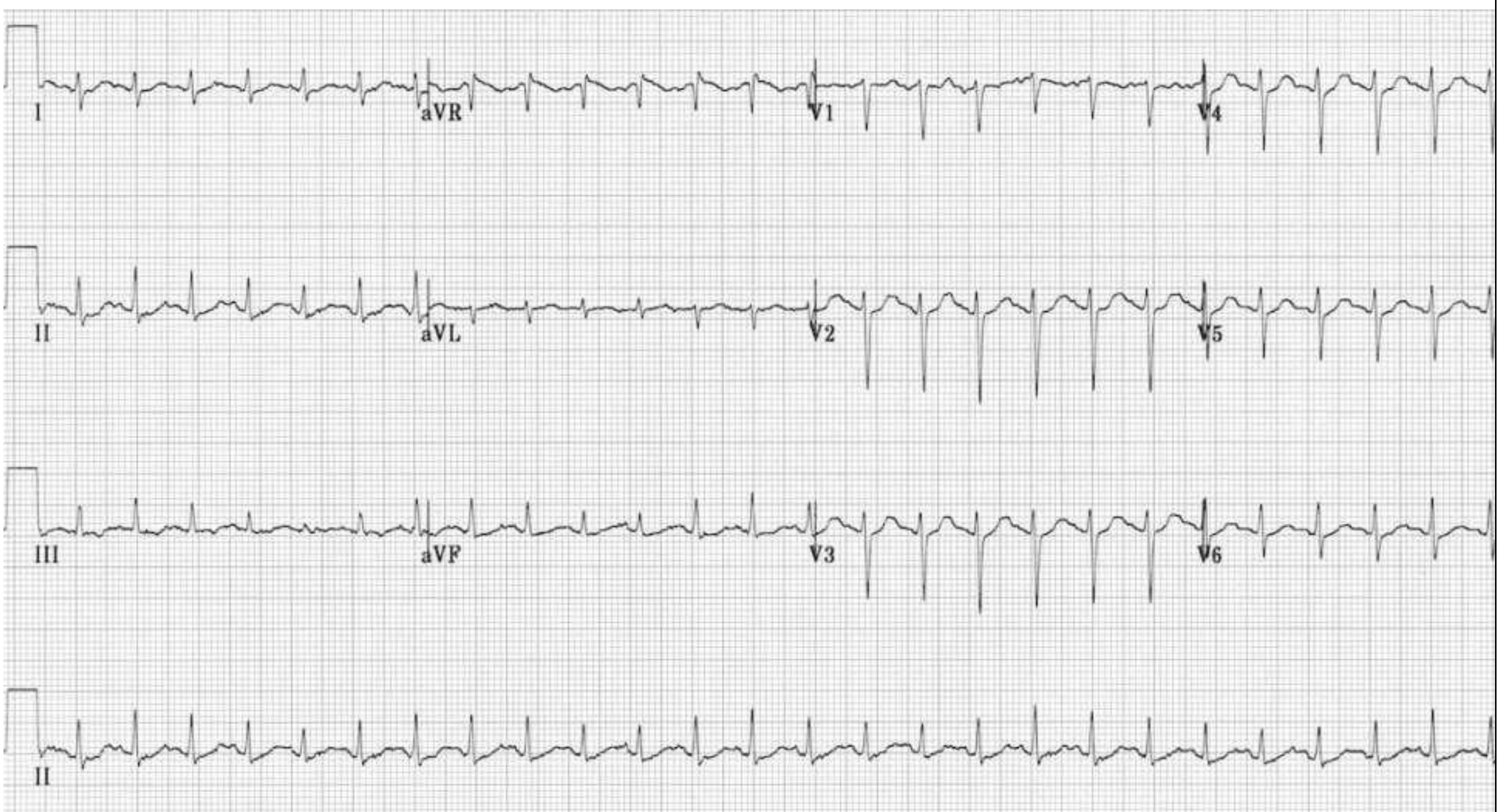
O2 Sat 96% GCS 14		reaching 36 – 38 in a few hours) ☐ Consult hospitalist/internal medicine	- End case after handover to hospitalist	

Appendix A: Laboratory Results

CBC WBC **14** Hgb **170** Plt **400** Lytes Na **147** K 4.5 Cl 102 HCO_3 **18** AG **14** Urea **12** Cr **280** Glucose 5.6 CK **2146** TSH Normal VBG pH **7.** pCO_2 **30** HCO_3 **18** Lactate **3.2**	Cardiac/Coags Trop 14 (hsTn) INR Normal aPTT Normal Biliary AST **1100** ALT **1200** GGT**400** ALP **320** Bili Normal Lipase Normal Tox EtOH 12 ASA Normal Tylenol Normal Osmols Normal

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



https://litfl.com/sinus-tachycardia-ecg-library/

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Important discussion points:

1. Exertional vs classic heat stroke (exertion related vs heat exposure related)
    a. Exertional – with exercise, more likely to be young, no comorbidities, sweaty, flushed skin
    b. Classic – comorbidities, older, higher mortality, not as diaphoretic
2. Treatment differences (Level 1A for cold water immersion for exertional vs 1C for classic usually due to elderly and comorbid nature of the latter)
3. Cold water immersion is 1st line for exertional heat stroke and can cool as rapidly as 0.2C/min
4. Target cooling temp of no less than 39C
5. Adjuncts – 4C IV fluids, fan and cold water to skin, ice packs to cover entire skin surface. If targeting specific areas, target palms, soles, cheeks.
6. Antipyretics are contraindicated in heat related illness – would not work
7. If cooling is not working – could intubate and paralyze
8. To prevent shivering – benzos, dexmedetomidine

References

1. Lipman et al. (2019) Wilderness Medical Society Clinical Practice Guidelines for the Prevention and Treatment of Heat Illness: 2019 Update. *Wilderness Medical Society Clinical Update Guidelines. 30*(4), S33 – S46. https://www.wemjournal.org/article/S1080-6032(18)30199-6/fulltext.

2. Platt, M.A., Price, T.M. (2023) Heat Illness. Rosen's Emergency Medicine: Concepts and Clinical Practice. (10e, Chapter 129) Elselvier Inc.